

Same Goal, Different Words: Spatial Instruction Sensitivity in World–Action Models

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Abstract: A robot should respond to the requested physical goal, not to one favored way of describing it. We test this requirement by asking three world–action models to perform the same placement while naming either the moved object or the reference object first in the spatial relation. Across 864 matched rollouts on four RoboLab scenes, reference-first descriptions produce fewer stable spatial outcomes, with an average gap of 12.2 percentage points. This gap is concentrated in particular relations rather than shared uniformly across scenes. We also find that reaching a spatial arrangement does not establish that it will be retained or that the intended object was moved. These findings motivate a more discriminating evaluation of world–action models: test alternative descriptions of the same goal, examine individual relations, and distinguish reaching a spatial arrangement from completing the instruction.

Keywords: World–Action Models, Spatial Language, Robot Evaluation

1 Introduction

Placing a mustard bottle to the right of a raisin box also leaves the box to the bottle’s left. Both descriptions specify the same arrangement, and both instructions can explicitly identify the bottle as the object to move. If behavior changes from the same starting state, physical task difficulty alone cannot explain the difference.

This is a useful test for world–action models, which generate actions alongside predictions of future observations. For robot control, predictive capability should be assessed alongside reliable goal conditioning. We ask: *when the physical goal and initial state stay fixed, does reversing the description of the spatial relation change what the model accomplishes?*

Instruction sensitivity is already established in robot learning. LIBERO-Plus evaluates language perturbations [1], LIBERO-Para tests meaning-preserving paraphrases from shared starts [2], and semantic guidance for world–action models targets instruction alignment [3]. Our contribution is a focused comparison of relational reference inversion across three world–action policies in existing RoboLab scenes [4]. As in controlled simulation evaluations [5], shared starts help separate the intervention from scene difficulty.



Same requested placement; same object to move

Mover-first (S): Move the mustard bottle on the table so that the mustard bottle is to the right of the raisin box.

Reference-first (I): Move the mustard bottle on the table so that the raisin box is to the left of the mustard bottle.

Figure 1: Exact tested mustard-right prompts and two exterior views of the asset from a pilot reset. Confirmation pairs share a frozen start; policies also receive a wrist view. The intervention changes argument order and relation words, not merely token order.

2 Paired spatial evaluation

Change the description, hold the goal fixed. We use four RoboLab scenes: a cube and bowl, butter and a raisin box, mustard and a raisin box, and two bowls. Goals cover left/right, front/behind, supported placement on top, and stacking either bowl on the other. Seven goals are native to their scenes; five are added conditions on existing assets. A planned two-bin scene failed scripted feasibility qualification and was excluded before confirmation, independently of learned-policy outcomes.

Every goal has three descriptions: direct wording (D), native where available; an expanded description that names the moved object first in the relation (S); and a description that names the reference object first and uses the converse relation (I). Figure 1 shows an exact pair. “Reference-first” refers to the spatial clause, not a change of viewpoint or of the object requested to move. For supported placement, “on top of and supported by” becomes “underneath and supporting.” Bowl wording is less exact: “underneath and supporting” does not explicitly require the nesting imposed by its scorer. Its results are shown separately.

Compare models from the same starts. We evaluate the DROID checkpoints of Cosmos3 Nano [6], Cosmos3 Edge [7], and FLUX 3 Action [8]. Inputs comprise the native wrist and two exterior cameras and proprioception. We use the first eight controller-qualified starts per scene under small position and yaw perturbations. Every model runs each goal and description from these starts: 864 confirmation episodes, excluding development runs. Paired initial-state and first-observation hashes match.

Policies execute 32-action chunks at 15 Hz, with fixed inference seed 6100 and checkpoint-specific inference settings. Episodes run for their full scene-specific horizon (20–40 s), without stopping at the first successful placement. Exact prompts, settings, states, and receipts accompany the paper. These prespecified secondary wording analyses became the paper’s focus after outcomes were available. At that stage, automated vision-language annotators replaced planned human annotations; poor agreement and no detectable diagnostic improvement left the original forecast-diagnosis question unresolved. The protocol and amendment accompany the artifacts.

Separate reaching a placement from completing the task. The prespecified measure checks whether the robot-frame relation, support, and gripper detachment hold for one continuous second while the designated mover remains below 0.02 m/s. This *stable spatial outcome* can occur anywhere in the episode. A post-hoc terminal check evaluates stability during the final second. Neither score verifies which object was moved: moving the reference may also make the relation true.

We identify goals already satisfied at reset from the frozen state registry. Cube-right, butter-right, and mustard-left are initially satisfied at every start. The prespecified wording contrasts include both initially satisfied and unsatisfied goals, averaging paired differences equally across models, scenes, and goals within scene. Initially satisfied goals contribute zero differences to the stable-outcome contrasts; the goal-level breakdown isolates new-goal achievement. Confidence intervals resample physical starts within each scene, keeping their associated models, goals, and descriptions together; paired cluster sign-flip tests use Holm correction for the two contrasts.

3 Results

Reference-first descriptions produce fewer stable outcomes. The mover-first form yields stable spatial outcomes more often than the reference-first form, with a weighted gap of 12.2 percentage points (95% interval: 8.6–15.4; adjusted $p < 0.001$). The average has the same sign for all three models (Figure 2), while the direct-versus-expanded contrast is smaller and inconclusive. The post-hoc terminal check retains a similar reference-inversion gap, so the finding is not explained solely by transient placements.

A few relations drive the average. The largest pooled gap is for mustard-right, with the same direction in all three models. The pooled descriptive gap remains positive when bowl stacking is

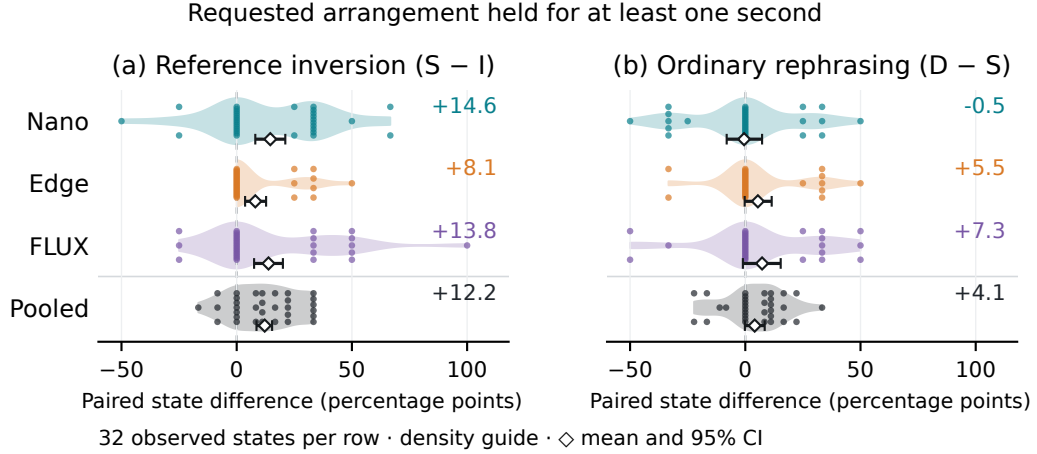


Figure 2: Paired wording effects. Positive S–I favors mover-first descriptions; D–S compares direct and expanded wording. Each dot averages paired differences over goals within one start (also over models in the pooled row). Shading guides the eye; diamonds and bars show means and 95% state-bootstrap intervals. Model-level intervals are descriptive.

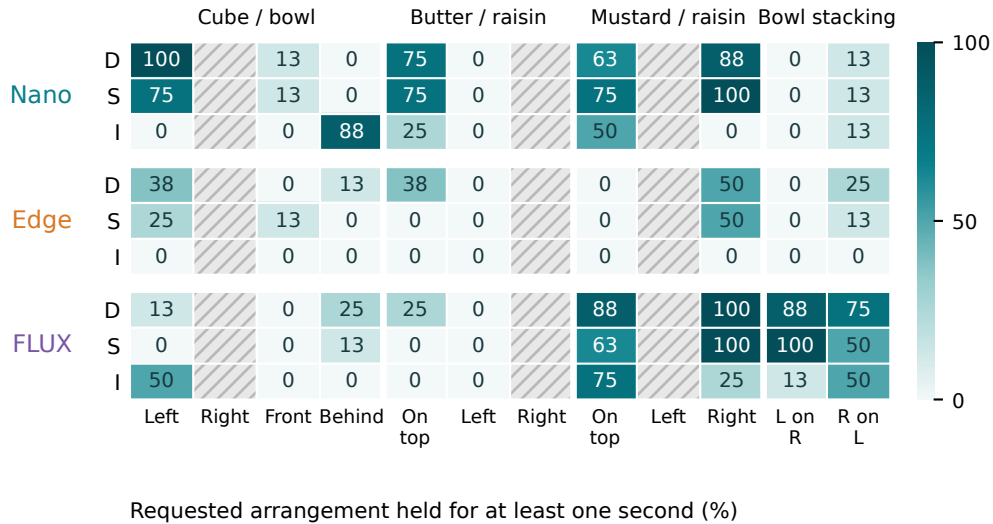


Figure 3: Reaching a stable spatial arrangement, by goal and wording. Each colored cell has eight starts; hatching marks initially satisfied goals. Relation passes do not verify complete instruction execution, and bowl wording has a support/containment ambiguity.

excluded. In contrast, the cube scene has no aggregate S/I difference, and cube-behind actually favors the reference-first form under the recorded metric. Figure 3 exposes these differences: performance varies with the particular scene, relation, and model. This does not establish a general inability to interpret spatial relations.

The success definition changes the interpretation. Roughly half of episodes with a stable outcome fail the terminal check. This measures retention under continued control, not performance with success-triggered stopping. Object roles also matter. All seven reference-first cube-behind passes come from Nano episodes with no detected cube lift and substantial bowl displacement. These summaries establish the relation, not that the cube was moved as instructed. A relation score alone cannot distinguish the requested manipulation from movement of the reference.

4 Discussion

An instruction specifies both a desired arrangement and an intervention: which object to move to produce it. Equivalent descriptions should preserve these requirements, even when successful trajectories differ. Our paired comparisons show that this consistency cannot be assumed. Shared starts rule out scene difficulty as the sole explanation for the wording gap, but cannot separate a misinterpreted goal from different object selection or execution of an otherwise correct plan.

Mustard-right and cube-behind show why an average is insufficient. Inversion reduces stable outcomes for the former yet improves the relation score for the latter, where passes coincide with substantial bowl displacement. Even a higher score need not establish better instruction following. Evaluation should distinguish whether the requested arrangement was reached, whether the intended manipulation was performed, and whether the arrangement was retained under continued control. These are distinct requirements for reliable behavior.

For world-action models, this raises a question beyond whether predicted futures look plausible: do prediction and action remain consistent with the intended intervention across descriptions? Our behavioral measurements cannot answer it. A useful next test would determine whether wording changes the predicted goal or object motion, and whether execution agrees with that prediction. This requires aligned predictions and executions, with vocabulary and argument order tested separately. Bowl support/containment ambiguity further limits equivalence claims. The aim is to locate failures within instruction following, rather than infer understanding from a relation score alone.

References

- [1] S. Fei, S. Wang, J. Shi, Z. Dai, J. Cai, P. Qian, L. Ji, X. He, S. Zhang, Z. Fei, J. Fu, J. Gong, and X. Qiu. LIBERO-Plus: A progressive robustness benchmark for visual-language-action models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 38574–38583, 2026.
- [2] C. Kim, M. Kim, M. Kang, H. Kim, and D. Jung. LIBERO-Para: A diagnostic benchmark and metrics for paraphrase robustness in VLA models. *arXiv preprint arXiv:2603.28301*, 2026. URL <https://arxiv.org/abs/2603.28301>. Accepted to EMNLP 2026, main conference.
- [3] J. He, J. Li, Z. Zhong, H. Yan, R. Li, Y. Zheng, J. Zhu, T. Zhang, Y. Du, W. Chen, S. Zhou, and H. Li. SG-WAM: Text-grounded and spatial-aware semantic guidance for world-action models. *arXiv preprint arXiv:2608.08839*, 2026. URL <https://arxiv.org/abs/2608.08839>.
- [4] X. Yang, R. Dagli, A. Zook, H. Hadfield, A. Goyal, S. Birchfield, F. Ramos, and J. Tremblay. RoboLab: A high-fidelity simulation benchmark for analysis of task generalist policies. In *Proceedings of Robotics: Science and Systems*, 2026. doi:10.15607/RSS.2026.XXII.096.
- [5] X. Li, K. Hsu, J. Gu, O. Mees, K. Pertsch, H. R. Walke, C. Fu, I. Lunawat, I. Sieh, S. Kirmani, S. Levine, J. Wu, C. Finn, H. Su, Q. Vuong, and T. Xiao. Evaluating real-world robot manipulation policies in simulation. In *Proceedings of the 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 3705–3728. PMLR, 2025.
- [6] NVIDIA. Cosmos3-Nano-Policy-DROID. Hugging Face model card and checkpoint, 2026. URL <https://huggingface.co/nvidia/Cosmos3-Nano-Policy-DROID>. Accessed September 27, 2026.
- [7] NVIDIA. Cosmos3-Edge-Policy-DROID. Hugging Face model card and checkpoint, 2026. URL <https://huggingface.co/nvidia/Cosmos3-Edge-Policy-DROID>. Accessed September 27, 2026.
- [8] Black Forest Labs. FLUX 3 Action DROID. Hugging Face model card and checkpoint, 2026. URL <https://huggingface.co/black-forest-labs/flux-3-action-droid>. Accessed September 27, 2026.